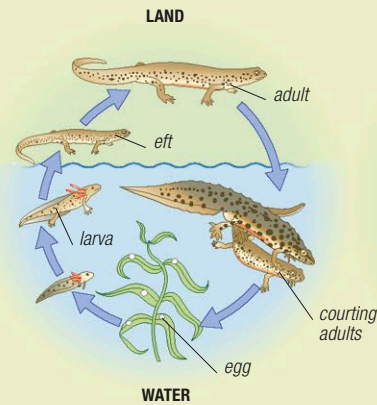


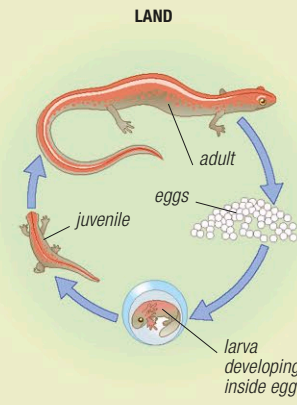
Life cycles

Newts and salamanders have complex life cycles, which typically involve 3 distinct stages: the egg, larva, and adult. The female lays eggs that contain yolk to support the growth of the developing embryo. Some species lay many small eggs, others a few large eggs, each containing a lot of yolk. The eggs hatch into larvae that have feathery external gills and feed on small animals. The larvae grow until they reach the stage at which they undergo metamorphosis into the adult form; this involves the loss of the gills and a switch to breathing air via the lungs and skin. The juvenile stage, which in some is known as the eft stage, is spent on land and may last for several years. In the adult stage, individuals become sexually mature and start to breed; this may involve a brief return to water. There are many variations on this basic pattern, which usually coincide with whether the species is amphibious, terrestrial, or aquatic (see right).



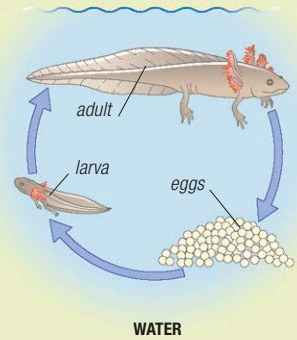
AMPHIBIOUS

This life cycle is typical of American, European, and Asian newts. Adults spend most of their life on land but return to water each spring to breed. This involves a partial reversal of metamorphosis, as individuals develop skin that can absorb oxygen from water and a tail like that of the larva, enabling aquatic newts to swim powerfully.



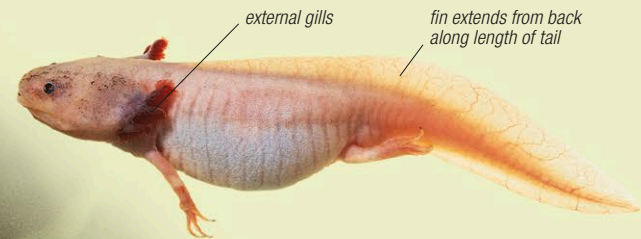
TERRESTRIAL

The eggs of terrestrial salamanders are typically laid on land. There are usually relatively few, large eggs, each containing a lot of yolk, which may be protected by the mother. The larval stage is completed within the egg, which hatches to produce a miniature terrestrial adult. In some species, the eggs may be retained inside the female's body. She gives birth, either to aquatic larvae or to terrestrial juveniles.



AQUATIC

In the aquatic life cycle, all the life stages are completed in water. The female typically produces a large number of small eggs and does not care for them. The adults become sexually mature while retaining many of the anatomical and physiological features of the larval stage. In many species, there are individual populations that have an aquatic life history, while others have an amphibious one.



NEOTENIC NEWTS AND SALAMANDERS

In neotenic newts and salamanders, individuals become sexually mature while still in the larval stage. They are fully aquatic and have large, feathery external gills.

The best known example is the axolotl (shown here), which can be artificially made to metamorphose into a terrestrial adult form by injecting it with the hormone thyroxine.

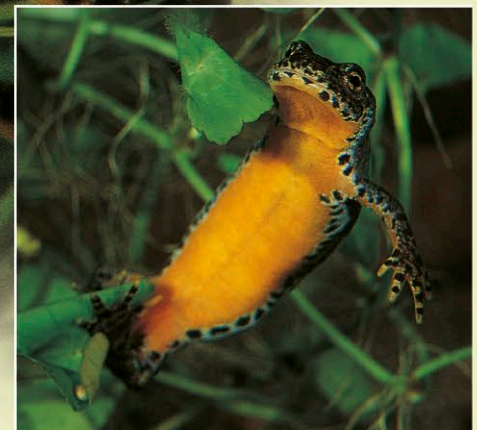
Defense

For most newts and salamanders, the risk of being attacked by a predator is greatly reduced by being active only at night. Many species also produce distasteful or toxic secretions from glands in their skin. These glands may be concentrated on the head or the tail. The skin secretions of Californian newts are particularly lethal. Such species are often brightly colored to warn predators, although the effectiveness of this coloration depends on predators learning to associate it with unpleasant consequences. A few species do not produce noxious secretions but mimic the color patterns of those that do. Several species are able to shed their tails when attacked. The detached tail, which twitches after it is dropped, distracts the attention of a predator away from the animal's body.



UNKEN POSITION

Some species of newts and salamanders adopt bizarre defensive positions. This Italian spectacled salamander has taken the contorted "unken" position to alarm predators and expose its bright warning coloration.



COLORATION

Many newts and salamanders have skin patterns and colors that provide camouflage. Others are very brightly colored, usually indicating their capacity to produce noxious or toxic skin secretions. However, in some species (such as this alpine newt), males become more brightly colored to attract females during the breeding season.